The Superconducting Phase Diagram of $V_3Si_{0.98}X_{0.02}$ with X=Mn, Fe, Co, Ni, Cu

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Measurements: Superconducting transitions, magneto-resistivity (H = 0 to 9 T) and dilatometry (H = 0 T) in the temperature range 2 – 300K.

Samples: Polycrystalline $V_3Si_{0.98}X_{0.02}$ where X = Mn, Fe, Co, Ni, Cu.

Motivation: The compound V_3Si shows a low-temperature martensite-like structural transformation at T_M ~ 21 K close to its T_{SC} ~ 17 K. We are investigating the relationship of T_M and T_{sc} upon doping of Period IV transition metals: Mn, Fe, Co, Ni, Cu.

(1) Introduction:

(a) V_3Si is known to have a structural phase transformation, from cubic to tetragonal, at T_M~ 21K, a few degrees above its superconducting temperature, T_{sc}~17K.[1]

(b) Doping V_3Si with a series of magnetic/non-magnetic dopants (Mn, Fe, Co, Ni, and Cu) across Period IV of the periodic chart may result in changes to T_M and T_{SC}.

(c) The superconducting phase diagram[2] of the parent compound V_3Si shown in Fig. 3 illustrates that T_{SC} -> 0 K at H ~ 21 T and T_M -> 0 K at H ~ 90 T.

(d) Based on data from dilatometry and magneto-resistivity measurements we are be able to determine the relationship between T_M and T_{SC} resulting from H-field and transition metal doping. A superconducting phase diagram H vs T_{SC} can be constructed from this investigation thereby illustrating changes from the parent compound behavior.

Manganese	Iron	Cobalt	Nickel	Copper
25	26	27	28	29
Mn	Fe	Co	Ni	Cu
Rad. = 124.0 pm	Rad. = 124.1 pm	Rad. = 124.3 pm	Rad. = 124.6 pm	Rad. = 127.8 pm
[Ar] $4s^2\ 3d^5$	[Ar] $4s^2\ 3d^6$	[Ar] $4s^2\ 3d^7$	[Ar] $4s^2\ 3d^8$	[Ar] $4s^1\ 3d^{10}$

Doping at. radii and electron concentration increase with increasing atomic number across Period IV.

(2) Magneto-Resistance and T_{SC} Transitions:

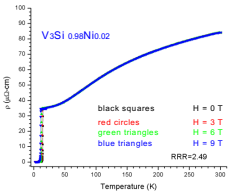


Fig. 1: As an example of our typical resistivity vs. temperature data, we show $V_3Si_{0.98}Ni_{0.02}$ with monotonic "good metal" behavior from 300 K down to ~ 50 K where we have evidence of a change in the scattering mechanism, upward curvature, and then on to the T_{SC} transition at T ~ 10 K. Data reveals no appreciable change in resistivity with application of magnetic field until T_{SC}.

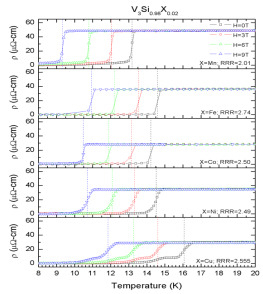


Fig. 2: Resistivity of $V_3Si_{0.98}X_{0.02}$ as a function of temperature and magnetic field obtained from AC transport measurements, where X = Mn, Fe, Co, Ni, Cu. Sample quality is compared by calculation of RRR at H = 0 T. Evidence of a double transition for Cu doping may indicate either a mixed phase or an inhomogeneous alloy.

(3) Field-dependent T_M and T_{SC}:

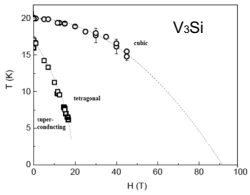
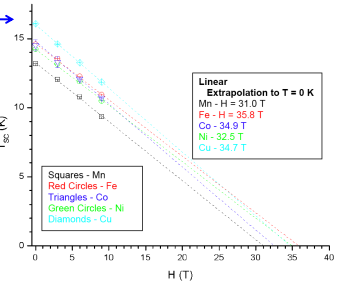


Fig. 3: Recent data by P. Goddard and J. C. Lashley[2] show the effect of H-field on T_M and T_{SC} of the parent compound V_3Si. Both resistivity and dilatometry measurements (not shown here) clearly indicate a trend in T_M and T_{SC} with magnetic field. This superconducting phase diagram reveals that as T_{SC} ➔ 0 K then H ➔ 20 T and as T_M ➔ 0 K then H ➔ 92 T.

Fig. 4: Superconducting transition temperatures vs. applied magnetic fields are compared for the five transition metal dopants. From a linear extrapolation we calculate the maximum H-field for the suppression of T_{SC} to 0 K.



(4) Dilatometry Measurements and dρ/dT vs. T:

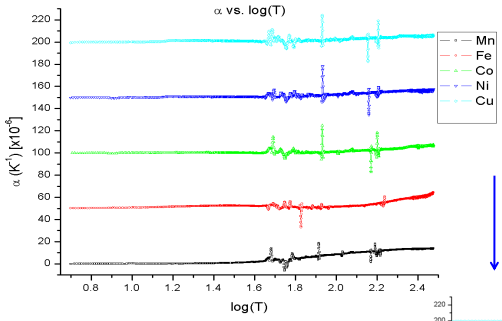
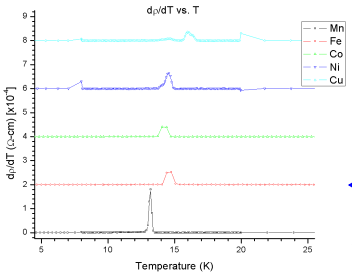


Fig. 5: Dilatometry measurements of the thermal expansion coefficient, α, for $V_3Si_{0.98}X_{0.02}$ as a function of log(T). The expansion coefficient for all doped compounds declines with decreasing T. No evidence of T_M is indicated by the data. The data illustrating different dopants has been offset by 5×10^{-5} (K^{-1}) for clarity of comparison.

Fig. 6: This plot accentuates the low temperature behavior of Fig. 5. The dashed vertical lines indicate the location of the superconducting transition temperatures for the various samples, whereas, the dark green dashed line to the right [log (1.32) K ~ 21 K] indicates the location of T_M for the parent compound V_3Si.



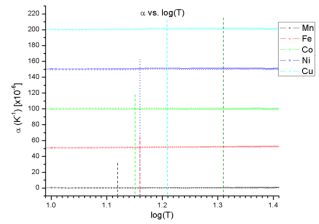


Fig. 7: Similar to the dilatometry data above, the plot of dρ/dT vs. T for the various dopants indicates no sudden shift from cubic to tetragonal crystal structure. The martensite transitions appear to have been washed out by the addition the transition metals, Mn, Fe, Co, Ni, Cu. The double transition in the Cu doped sample is evident here.

(5) Discussion & Conclusions:

(1) All the dopants utilized here are found to slightly alter the superconducting transition temperature. Fig. 2 shows the three ferromagnetic dopants (Fe, Co and Ni) shift the T_{SC} by ~ -2.5 K. Where as the paramagnetic dopant, Mn, drives T_{SC} down by ~ 4 K, and diamagnetic-doped, Cu, decreases T_{SC} by ~1 K. Since both atomic radii and electron number increase across Period IV the trend in T_{SC} correlates well with this series of dopants.

(2) Evidence from both dilatometry, Fig. 5 and resistivity Figs. 2, 7 shows that the martensitic transition is no longer present in the neighborhood of 21 K as a result from the transition metal doping. Doping at the 2% level appears to prevent or suppress the cubic to tetragonal martensite transition.

(6) Future Work:

(1) Resistance measurements to complete the SC phase diagram for these doped V_3Si compounds is planned for T_{SC} -> 0 K and H > 9 T at the NHMFL.

(2) Specific heat measurements below T = 50 K are planned as part of this analysis at LANL to indicate any possible thermodynamic interactions between T_M and T_{SC}.

(7) References:

[1] B. W. Batterman and C. S. Barrett, Phys. Rev. **145** 206 (1966). M. Liu and T. R. Finlayson, Phys. Rev. B **52**, 530 (1995).

[2] X. Yang, *et al.*, Phil. Mag. *(accepted for publication).*